

Bandhav Veluri

bandhav@cs.washington.edu

<https://homes.cs.washington.edu/~bandhav/>

RESEARCH INTERESTS

Speech/Audio Language Modeling, Conversational AI and Human-Computer Interaction

EDUCATION

University of Washington, Seattle, WA Jul 2020 to Jun 2025
Ph.D. in Computer Science and Engineering
Advisors: Shyam Gollakota and Michael Taylor

University of Washington, Seattle, WA, Jun 2020
M.S. in Electrical and Computer Engineering

Indian Institute of Technology Roorkee, Roorkee, India, May 2016
Bachelor of Technology in Electrical Engineering

WORK EXPERIENCE

Sesame AI, Bellevue, WA Sep 2025 to Present
Research Scientist

Meta, Seattle, WA May 2025 to Sep 2025
Research Scientist

Meta, Seattle, WA Sep 2023 to Mar 2025
Visiting Researcher

Texas Instruments, Bangalore, India Jul 2016 to Jul 2018
Digital Signal Processing Engineer

Janacare, Bangalore, India May 2015 to Jul 2015
Embedded Software Intern

SELECT PUBLICATIONS

AV-Dialog: Spoken Dialogue Models with Audio-Visual Input
*Tuochao Chen, **Bandhav Veluri**, Hongyu Gong, Shyamnath Gollakota*
Association for Computational Linguistics (ACL Main 2026)

Beyond Turn-Based Interfaces: Synchronous LLMs as Full-Duplex Dialogue Agents
***Bandhav Veluri**, Benjamin Peloquin, Bokai Yu, Hongyu Gong, Shyamnath Gollakota*
Conference on Empirical Methods in Natural Language Processing (EMNLP Main 2024)

IRIS: Wireless Ring for Vision-based Smart Home Interaction
Maruchi Kim, Antonio Glenn*, **Bandhav Veluri***, Yunseo Lee, Eyoel Gebre, Aditya Bagaria, Shwetak Patel, Shyamnath Gollakota* (*equal contribution)
ACM Symposium on User Interface Software and Technology (UIST 2024)

Look Once to Hear: Target Speech Hearing with Noisy Examples
Bandhav Veluri, Malek Itani*, Tuochao Chen, Takuya Yoshioka, Shyamnath Gollakota* (*equal contribution)
ACM Conference on Human Factors in Computing Systems (CHI 2024)

Scalable, Programmable and Dense: The HammerBlade Open-Source RISC-V Manycore
*Dai Cheol Jung, Max Ruttenberg, Paul Gao, Scott Davidson, Daniel Petrisko, Kangli Li, Aditya K Kamath, Lin Cheng, Shaolin Xie, Peitian Pan, Zhongyuan Zhao, Zichao Yue, **Bandhav Veluri**, Sripathi Muralitharan, Adrian Sampson, Andrew Lumsdaine, Zhiru Zhang,*

Christopher Batten, Mark Oskin, Dustin Richmond, Michael Bedford Taylor
International Symposium on Computer Architecture (ISCA 2024)

Semantic Hearing: Programming Acoustic Scenes with Binaural Hearables
*Bandhav Veluri**, *Malek Itani**, *Justin Chan, Takuya Yoshioka, Shyamnath Gollakota*
(*equal contribution)
ACM Symposium on User Interface Software and Technology (UIST 2023)

REAL-TIME TARGET SOUND EXTRACTION
Bandhav Veluri, Justin Chan, Malek Itani, Tuochao Chen, Takuya Yoshioka, Shyamnath Gollakota
IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP 2023)

NeuriCam: Key-Frame Video Super-Resolution and Colorization for IoT Cameras
Bandhav Veluri, Ali Saffari, Collin Pernu, Joshua Smith, Michael Taylor, Shyamnath Gollakota
Annual International Conference On Mobile Computing And Networking (MobiCom 2023)

A Tensor Processing Framework for CPU-Manycore Heterogeneous Systems
Lin Cheng, Peitian Pan, Zhongyuan Zhao, Krithik Ranjan, Jack Weber, Bandhav Veluri, Seyed Bornha Ehsani, Max Ruttenberg, Dai Cheol Jung, Preslav Ivanov, Dustin Richmond, Michael B Taylor, Zhiru Zhang, Christopher Batten
IEEE TCAD 2021

BlackParrot: An Agile Open-Source RISC-V Multicore for Accelerator SoCs
Daniel Petrisko, Farzam Gilani, Mark Wyse, Dai Cheol Jung, Scott Davidson, Paul Gao, Chun Zhao, Bandhav Veluri, Tavio Guarino, Zahra Azad, Sadullah Canakci, Ajay Joshi, Mark Oskin and Michael Bedford Taylor
IEEE Micro 2020

The Celerity Open-Source 511-Core RISC-V Tiered Accelerator Fabric
Scott Davidson, Shaolin Xie, Christopher Torng, Khalid Al-Hawaj, Austin Rovinski, Tutu Ajayi, Luis Vega, Chun Zhao, Ritchie Zhao, Steve Dai, Aporva Amarnath, Bandhav Veluri, Paul Gao, Anuj Rao, Gai Liu, Rajesh K. Gupta, Zhiru Zhang, Ronald G. Dreslinski, Christopher Batten, Michael Bedford Taylor
IEEE Micro 2018

AWARDS

Best paper honorable mention for "Look Once to Hear" at CHI 2024

Recipient of the Meta AI Mentorship (AIM) Scholarship 2023-24.

Recipient of the Paul G. Allen School First-Year Ph.D. Fellowship 2020-21.

Recipient of the Merit-Cum-Means Scholarship at IIT Roorkee 2013-16.

Ranked 1556 among half a million students in Indian Institute of Technology Joint Entrance Examination (IITJEE) 2012.

MEDIA COVERAGE

Noise-canceling headphones use AI to let a single voice through
MIT Technology Review, May 23, 2024
Related: US News, UW News, GeekWire

Noise-canceling headphones could let you pick and choose the sounds you want to hear
MIT Technology Review, Nov 9, 2023
Related: UW News, New Atlas

**OPEN SOURCE
SOFTWARE**

LookOnceToHear (★ 597, 📄 66)

A novel human-interaction method for enhancing target speaker in real-time.

Waveformer (★ 324, 📄 34)

A deep neural network architecture for low-latency audio processing.

NeuriCam (★ 95, 📄 9)

Deep learning based video sensing method for low-power IoT cameras.